

GEN_DELAY

Overview

gen_delay allows users to insert delay on a single-bit or multi-bit signal. Delay (in clock cycles) is defined as a random value within a predefined range [MAX_CLOCKS: MIN_CLOCKS]. In order to specify an exact delay value, both parameters have to be the same.

Syntax

```
gen_delay #(min_clocks, max_clocks, bit_width) DelayInst (clk, rst_n,  
data_in, data_out);
```

Parameters	
min_clocks	Minimum limit of delay range
max_clocks	Maximum limit of delay range
bit_width	Data bit width. Default: 1
Interface	
Clk	Sampling clock for data
rst_n	Active-low reset
data_in	Input data signal. Bit width of this signal is defined with a "bit_width" parameter.
data_out	Output data signal, which equals to the delayed input one.

Verilog example

The current example shows how an 8-bit data_in signal is delayed by the random number of clock cycles [from 3 up to 6]

```
module test;  
  reg clk;  
  
  reg [7:0] data_in;  
  wire [7:0] data_out;  
  integer i;  
  
  always #15 clk <= ~clk;  
  gen_delay #(3,6,8) delay (clk, 1'b1, data_in, data_out);  
  
  initial begin  
    $dumpvars;  
    clk = 1;  
    data_in = 0;  
  
    for (i=0; i<200; i=i+1) begin  
      @(posedge clk);  
      if (i%10 == 0) data_in = i;  
      else data_in = 0;  
    end  
    #100 $finish;  
  end  
endmodule
```